

Necessity of the Analytic–Injective Criterion for Finite-Rank Spectral Densities

Independent research note

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Abstract

Dritschel–Rovnyak ask whether their analytic–injective multiplier criterion for factorability is necessary, recording only dimensions one and two as known. We prove necessity whenever the outer spectral factor has finite-dimensional output. This settles every finite-dimensional Hilbert space and, more generally, every factorable operator-valued function of uniformly finite rank on an arbitrary separable Hilbert space. The proof factors the finite-dimensional co-Gramian and then uses equality of Gram operators to construct a measurable partial isometry. We also isolate the precise analytic-minorant problem left in infinite dimension.

Packet status

Run	fa_banach_001, lane 13.
Source	arXiv:0903.3639, Theorem 4.3 and the question on PDF page 14 [1].
Classification	Candidate strong partial result, likely valid subject to human review.
Novelty	Moderate. No later explicit answer or all-finite-dimensional extension was located.

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evidently unpublished, but versions were given by D. Z. Arov [1] Lemma to Theorem 4] and A. S. Markus [41] Theorem 34.3 on p. 199]. Theorem 4.3 includes all of these versions.

We do not know if the conditions (i) and (ii) in Theorem 4.3 are necessary for factorability. It is not hard to see that they are necessary in the simple cases $\dim \mathfrak{G} = 1$ and $\dim \mathfrak{G} = 2$ (for the latter case, one can use [50] Example 1, p. 125]). The general case, however, is open.

Proof of Theorem 4.3. Let F satisfy (i) and (ii). Define a subspace $\mathfrak{M}$ of $L^2_{\mathfrak{G}}$ by

$$\mathfrak{M} = \bigcap_{n=0}^{\infty} \zeta^n \overline{F^{\frac{1}{2}} H_{\mathfrak{G}}^2} = \bigcap_{n=0}^{\infty} \overline{\zeta^n F^{\frac{1}{2}} H_{\mathfrak{G}}^2}.$$

1 Statement

Let $F \in L^{\infty}_{\mathcal{B}(\mathcal{H})}(\mathbb{T})$ be positive. Following the source, F is *factorable* if

$$F(\zeta) = A(\zeta)^* A(\zeta) \quad \text{a.e.}$$

for an operator-valued Smirnov function A ; the factor may be chosen outer. Because F is bounded, such an A belongs to $H_{B(\mathcal{H})}^\infty$.

Theorem 1 (finite-output necessity). *Suppose $F = A^*A$ with A outer and*

$$\overline{AH_{\mathcal{H}}^2} = H_{\mathcal{E}}^2$$

for a finite-dimensional subspace $\mathcal{E} \subseteq \mathcal{H}$. Then there is a weakly measurable $\psi \in L_{B(\mathcal{H})}^\infty$ with

- (i) $\psi F \in H_{B(\mathcal{H})}^\infty$;
- (ii) $\psi(\zeta)$ is one-to-one on $\overline{F(\zeta)\mathcal{H}}$ for almost every ζ .

Moreover $\|\psi\|_\infty \leq 1$, and $\psi(\zeta)$ is isometric on $\overline{F(\zeta)\mathcal{H}}$ almost everywhere.

Theorem 2 (two consequences). *The conditions in Theorem 4.3 of the source are necessary for factorability in each of the following cases:*

- (a) $\dim \mathcal{H} < \infty$;
- (b) $\mathcal{H}$ is arbitrary and $\text{rank } F(\zeta) \leq r < \infty$ almost everywhere.

Thus the dimensions one and two recorded in the source extend to every finite dimension.

For reference, the source's sufficient criterion is reproduced below.

We obtain a very useful sufficient condition for factorability.

Theorem 4.3. *Suppose $F \in L_{\Sigma(\mathfrak{G})}^\infty$ and $F \geq 0$ a.e. For F to be factorable, it is sufficient that there exists a function ψ in $L_{\Sigma(\mathfrak{G})}^\infty$ such that*

- (i) $\psi F \in H_{\Sigma(\mathfrak{G})}^\infty$;
- (ii) *for all $\zeta \in \mathbb{T}$ except at most a set of measure zero, $\psi(\zeta)|\overline{F(\zeta)\mathfrak{G}}$ is one-to-one.*

*If these conditions are met and $F = G^*G$ a.e. with G outer, then $\psi G^* \in H_{\Sigma(\mathfrak{G})}^\infty$.*

Theorem 4.3 appears in Rosenblum [49] with $\psi(\zeta) = \zeta^m$ (viewed as an operator-valued function). The case of an arbitrary inner function was proved and applied in a variety of ways by Rosenblum and Rovnyak [50, 51]. V. I. Matsaev first showed that more general functions ψ can be used. Matsaev's result is

2 Factoring the co-Gramian

We first separate the operator-theoretic step from the determinant argument.

Proposition 1 (co-Gramian principle). *Let $F = A^*A$ with $A \in H_{B(\mathcal{H})}^\infty$. Suppose, on a fixed closed subspace $\mathcal{E}$ containing the ranges of A , that*

$$AA^* = Q^*Q \quad \text{a.e.}$$

for $Q \in H_{B(\mathcal{E})}^\infty$. Then the conclusion of Theorem 1 holds.

Proof. At almost every boundary point the maps

$$A^* : \mathcal{E} \longrightarrow \mathcal{H}, \quad Q : \mathcal{E} \longrightarrow \mathcal{E}$$

have the same Gram operator:

$$(A^*)^* A^* = A A^* = Q^* Q.$$

Therefore the rule $U(A^* y) = Q y$ defines an isometry from $\text{ran } A^*$ to $\text{ran } Q$. Extend it continuously to $\overline{\text{ran } A^*}$ and by zero on its orthogonal complement. This gives a partial isometry $U : \mathcal{H} \rightarrow \mathcal{E}$ satisfying $U A^* = Q$. The standard measurable polar decomposition makes $U(\zeta)$ weakly measurable; in finite dimensions it is equivalently obtained using the measurable Moore–Penrose inverse.

Let $J : \mathcal{E} \hookrightarrow \mathcal{H}$ be the inclusion and put $\psi = J U$. Then $\|\psi\| \leq 1$, it is isometric on $\overline{\text{ran } A^*}$, and

$$\psi F = J U A^* A = J Q A \in H_{\mathcal{B}(\mathcal{H})}^\infty.$$

Finally $\overline{\text{ran } F} = \overline{\text{ran } A^*}$ because the two spaces have the same orthogonal complement $\ker A$. This proves both required conditions. $\square$

Now assume $r = \dim \mathcal{E} > 0$ in Theorem 1. Outerness implies

$$\overline{\text{ran } A(0)} = \mathcal{E}. \tag{1}$$

Indeed, approximate any constant function $e \in H_{\mathcal{E}}^2$ by functions $A f_n$ and apply the bounded evaluation map at zero. Since $\mathcal{E}$ is finite dimensional, $A(0)$ is onto.

Choose an isometry $V : \mathbb{C}^r \rightarrow \mathcal{H}$ such that $B(0) := A(0)V : \mathbb{C}^r \rightarrow \mathcal{E}$ is invertible, and set

$$W = A A^*, \quad B = A V, \quad \Delta = \det B.$$

The scalar function $\Delta \in H^\infty$ is nonzero. Since $V V^* \leq I$,

$$W = A A^* \geq A V V^* A^* = B B^*, \quad \det W \geq |\Delta|^2 \quad \text{a.e.} \tag{2}$$

Here determinant monotonicity is applied to positive $r \times r$ matrices. The boundary values of a nonzero bounded analytic scalar function are nonzero almost everywhere and $\log |\Delta| \in L^1(\mathbb{T})$ after separating positive and negative parts (Jensen's theorem). Inequality (2), together with the boundedness of W , yields

$$\log \det W \in L^1(\mathbb{T}), \quad W(\zeta) > 0 \quad \text{a.e.}$$

The matrix Szegő theorem quoted as Theorem 4.7 in the source now gives $Q \in H_{\mathcal{B}(\mathcal{E})}^\infty$ such that $W = Q^* Q$. (Boundedness follows from $\|Q\|^2 = \|W\|$ on the boundary.) Proposition 1 proves Theorem 1.

Theorem 4.7. *Suppose that F is an $r \times r$ measurable matrix-valued function having invertible values on $\mathbb{T}$ such that $F \geq 0$ a.e. and $\log^+ \|F\|$ is integrable. Then F is factorable if and only if $\log \det F$ is integrable.*

3 Finite rank and the remaining problem

Case (a) of Theorem 2 is immediate from Theorem 1. For (b), boundary rank satisfies $\text{rank } A(\zeta) = \text{rank } A(\zeta)^* A(\zeta) \leq r$ almost everywhere. Given any $r + 1$ domain vectors and $r + 1$ output vectors, the corresponding scalar minor of $A(z)$ is bounded analytic and has zero boundary values almost everywhere; it therefore vanishes identically. Hence $\text{rank } A(z) \leq r$ throughout the disk. Equation (1) then gives $\dim \mathcal{E} \leq r$, so Theorem 1 applies.

The argument also gives an exact gateway for the unsolved case. Necessity is equivalent to the existence of an analytic K such that, almost everywhere,

$$K^* K \leq C A A^*, \quad \ker K = \ker A^*. \quad (3)$$

Indeed, Douglas' lemma turns (3) into $K = \psi A^*$; the kernel identity supplies injectivity, and $\psi F = K A$ is analytic. The finite-dimensional determinant constructs $K = Q$. In infinite dimension, $A A^* + \varepsilon I$ regularization, compact smoothing, finite compressions, weighted direct sums, and iteration of the source's Sarason factorization were all audited. Each loses the kernel equality or leaves uncontrolled off-diagonal tails. Conversely, a nonfactorable $A A^*$ is not yet a counterexample, since (3) need not dominate $A A^*$ from below.

A bounded literature search on August 17, 2026 covered the exact question, later citations, and the phrases “necessary for factorability”, “ ψF ”, “ ψG^* ”, and finite-dimensional spectral factorization. It found no later explicit answer or statement of the extension above. The result should therefore be read as a candidate new partial answer, not as a claim that the unrestricted question is settled.

References

- [1] M. A. Dritschel and J. Rovnyak, *The Operator Fejér–Riesz Theorem*, arXiv:0903.3639 (2009).